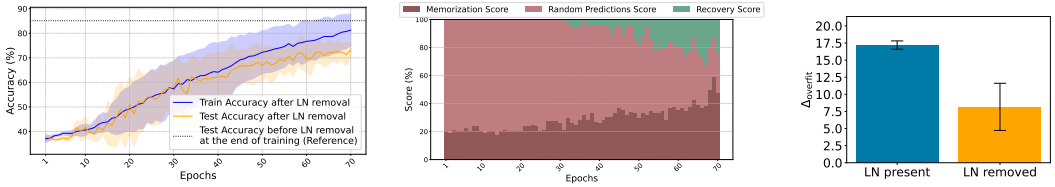
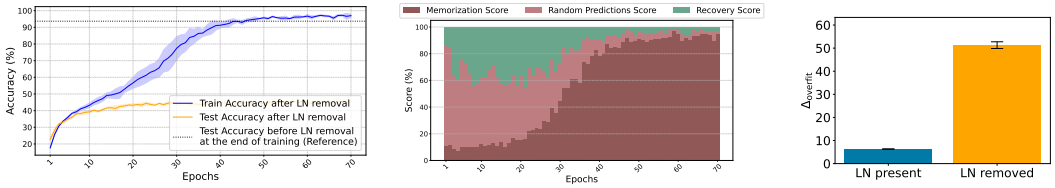


(a) Learning (Test) accuracy over epochs for Pre-LN Model (GPT2)

(b) Memorization, Recovery and Random Predictions over epochs for Pre-LN Model (GPT2)

(c) Overfitting gap for Pre-LN Model (GPT2)

Figure 7: **LN removal destabilizes learning in Pre-LN model - GPT2, TweetTopic Dataset:** LN removal critically affects learning while memorization still persists in GPT2. For GPT2, the overfitting gap decreased after LN removal, because the model could not even stabilize during training due to the destabilization of learning. Hence both train and test accuracies remain low and comparable. However, the learning accuracy still remains low when LN is absent in comparison to when LN was present, and struggle with high memorization and random predictions (red-color family bars).

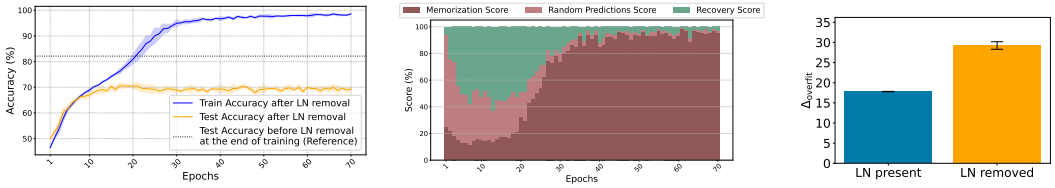


(a) Learning (Test) accuracy over epochs for Pre-LN Model (ViT-B)

(b) Memorization, Recovery and Random Predictions over epochs for Pre-LN Model (ViT-B)

(c) Overfitting gap for Pre-LN Model (ViT-B)

Figure 8: **LN removal destabilizes learning in Pre-LN model - ViT-B, CIFAR10 Dataset:** LN removal critically affects learning while memorization still persists in ViT-B. This further exacerbates overfitting seen by increasing train-test accuracy gap when LN is removed, due to the drop in test-accuracy.

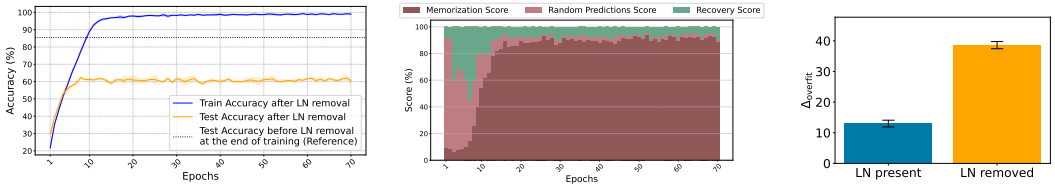


(a) Learning (Test) accuracy over epochs for Pre-LN Model (DeiT)

(b) Memorization, Recovery and Random Predictions over epochs for Pre-LN Model (DeiT)

(c) Overfitting gap for Pre-LN Model (DeiT)

Figure 9: **LN removal destabilizes learning in Pre-LN model - DeiT, UTK-Face Dataset:** LN removal critically affects learning while memorization still persists in DeiT. This further exacerbates overfitting seen by increasing train-test accuracy gap when LN is removed, due to the drop in test-accuracy.

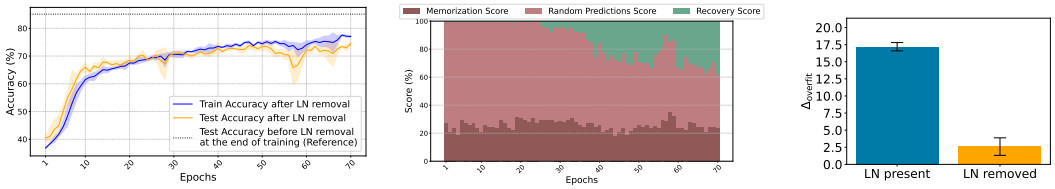


(a) Learning (Test) accuracy over epochs for Pre-LN Model (ViT-S)

(b) Memorization, Recovery and Random Predictions over epochs for Pre-LN Model (ViT-S)

(c) Overfitting gap for Pre-LN Model (ViT-S)

Figure 10: **LN removal destabilizes learning in Pre-LN model - ViT-S, NICO++ Dataset:** LN removal critically affects learning while memorization still persists in ViT-S. This further exacerbates overfitting seen by increasing train-test accuracy gap when LN is removed, due to the drop in test-accuracy.



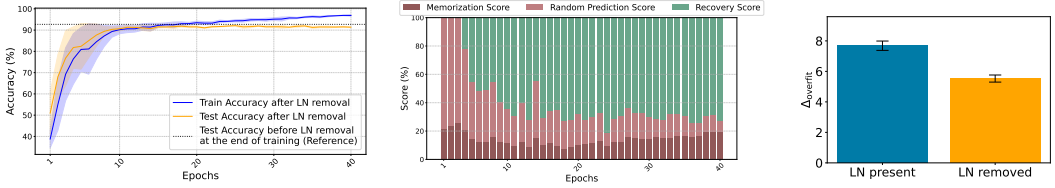
(a) Learning (Test) accuracy over epochs for Pre-LN Model (RoBERTa-PreLayerNorm)

(b) Memorization, Recovery and Random Predictions over epochs for Pre-LN Model (RoBERTa-PreLayerNorm)

(c) Overfitting gap for Pre-LN Model (RoBERTa-PreLayerNorm)

Figure 11: **LN removal destabilizes learning in Pre-LN model - RoBERTa-PreLayerNorm, TweetTopic Dataset:** LN removal critically affects learning while memorization still persists in RoBERTa-PreLayerNorm. For RoBERTa-PreLayerNorm, the overfitting gap decreased after LN removal, because the model could not even stabilize during training due to the destabilization of learning. Hence both train and test accuracies remain low and comparable. However, the learning accuracy still remains low when LN is absent in comparison to when LN is present, and struggles with high memorization and random predictions (red-color family bars).

G.1.2 Post-LN models - Suppression of Memorization & True Labels Recovery

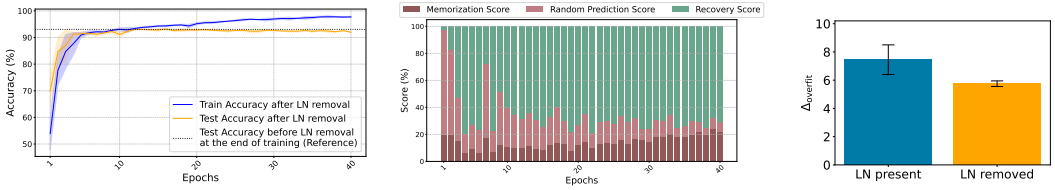


(a) Learning (Test) accuracy over epochs for Post-LN Model (BERT)

(b) Memorization, Recovery and Random Predictions over epochs for Post-LN Model (BERT)

(c) Overfitting gap for Post-LN Model (BERT)

Figure 12: **LN removal suppresses memorization & facilitates true label recovery in Post-LN model - BERT, Emotions Dataset:** LN removal in BERT suppresses memorization and facilitates true label recovery, while keeping learning intact. This further reduces overfitting seen by decreasing train-test accuracy gap when LN is removed.

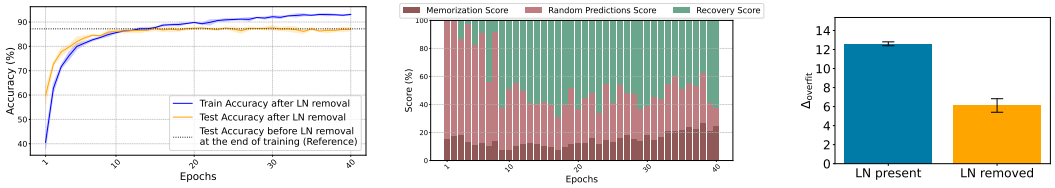


(a) Learning (Test) accuracy over epochs for Post-LN Model (DeBERTa)

(b) Memorization, Recovery and Random Predictions over epochs for Post-LN Model (DeBERTa)

(c) Overfitting gap for Post-LN Model (DeBERTa)

Figure 13: **LN removal suppresses memorization & facilitates true label recovery in Post-LN model - DeBERTa, Emotions Dataset:** LN removal in DeBERTa suppresses memorization and facilitates true label recovery, while keeping learning intact. This further reduces overfitting seen by decreasing train-test accuracy gap when LN is removed.



(a) Learning (Test) accuracy over epochs for Post-LN Model (Longformer)

(b) Memorization, Recovery and Random Predictions over epochs for Post-LN Model (Longformer)

(c) Overfitting gap for Post-LN Model (Longformer)

Figure 14: **LN removal suppresses memorization & facilitates true label recovery in Post-LN model - Longformer, News Dataset:** LN removal in Longformer suppresses memorization and facilitates true label recovery, while keeping learning intact. This further reduces overfitting seen by decreasing train-test accuracy gap when LN is removed.